PowerPulse

Electricity Demand and Generation Data Warehouse

Executive summary

PowerPulse organizes electricity demand, the publisher's day-ahead forecast and reported net generation by energy source. Planners can compare load patterns, analysts can review forecast performance, and managers can inspect changes in the generation mix through consistent historical queries.

The design follows a downloaded and profiled EIA-930 baseline. Demand and generation use separate grains, shared calendar and authority dimensions, explicit source versions and rules for missing values and changing fuel categories.

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Academic record	Details
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Institution	FAST NUCES Lahore
Course	Data Warehouse and Business Intelligence Lab (DL3004)
Phase	1 - Business Analysis and Dimensional Design

Data already obtained

87,720 authority-hour source records. 727,720 numeric reported fuel observations available for unpivoting. Five authorities. Four official CSVs. Two complete operating years.

Phase 1 contains the completed design and measured source evidence. The source acquisition and profiling scripts were executed. PostgreSQL warehouse loading, ETL execution and dashboards belong to later implementation.

Planning questions from hourly data

The warehouse supports historical analysis of demand, forecast performance and recorded generation.

Business problem

A large hourly feed is difficult to compare consistently when dates, source categories and missing values change. A joined table can also repeat the same demand value once for every fuel. PowerPulse defines a stable comparison grain and reusable metrics so planners can inspect trends and analysts can trace the numbers to published records.

Stakeholder	Decision supported
Electricity planners	Review peak demand, seasonal load and the supply mix in high-demand hours.
Operations analysts	Compare day-ahead forecasts with the corresponding reported actual values.
Energy analysts	Track source-specific net generation while accounting for changes in source categories.
Data steward / project team	Inspect missing measures, revisions and discrepancies between fuel sums and published totals.

Scope

Five U.S. balancing authorities (CISO, ERCO, ISNE, NYIS and PJM), with publisher operating dates from 1 January 2024 through 31 December 2025. The acquired baseline contains 87,720 authority-hour records over 731 operating dates.

The pilot covers demand profiles, forecast error, net generation categories and source-quality diagnostics. Future delivery is a PostgreSQL dimensional warehouse with reporting views and Power BI dashboards. The system supports analysis of bulk-grid operating areas. Generator control, consumer billing, household supply routing, a new forecasting model and emissions estimates are outside this scope.

Success criteria

Every question maps to observed columns, every fact has a declared grain, and every measure has an eligibility rule. The schema and dictionary agree, source files can be acquired again, and the raw evidence is sufficient for a reviewer to inspect a row and its provenance.

A downloaded two-year baseline

All counts on this page were measured from the included CSV extracts.

Observation	Measured result
Acquired original files	4 EIA-930 balance CSVs; 1,067,976 source rows before area filtering.
Selected baseline	87,720 records; 17,544 distinct UTC hours per authority; 731 operating dates.
Forecast scoring candidates	87,399 pairs satisfy actual > 0 and forecast >= 0 with numeric values.
Coverage limitations	218 blank values in each core measure. Fuel data includes negative net values and different column sets across files.
Identity	0 duplicate authority/UTC-hour pairs. 0 missing UTC hours within each authority’s observed span.

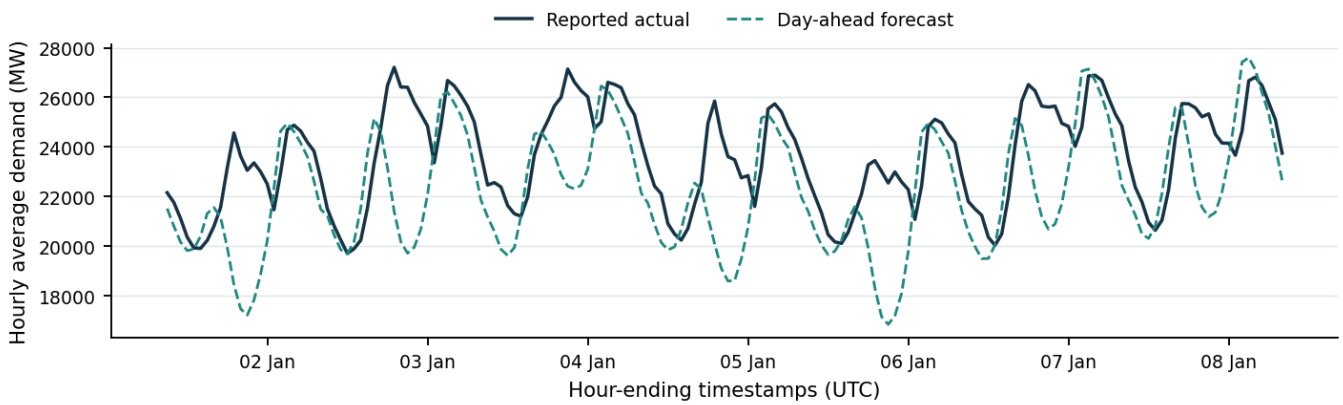


Figure 1. Actual records from CISO, operating dates 1-7 January 2025. The horizontal axis uses UTC hour endings. This figure illustrates the acquired source; no warehouse has been loaded.

Use the official bulk CSV route for the baseline. It supplies both business processes in one source record, keeps raw headers visible and requires no API key. The EIA reference workbook supplies names and region labels. The API is a possible later acquisition optimization after its units, timestamps and category crosswalk are tested against the bulk baseline.

Ten questions that the data supports

Use the business question document for field-level mappings and expected outputs.

ID	Analytical question
Q01	How does monthly demand energy change across the five operating areas?
Q02	Which local hours have the highest average demand?
Q03	When did each authority reach its monthly peak demand?
Q04	How do weekday and weekend demand patterns differ across seasons?
Q05	Which operating areas and months have the largest forecast errors?
Q06	When do forecasts systematically fall below actual demand?
Q07	How does reported generation by energy source change each month?
Q08	What share of reported net generation comes from wind and solar categories?
Q09	Does the generation mix differ between high-demand and other hours?
Q10	Where do missing measures and generation discrepancies affect reporting?

Population and reporting time

Use the five selected authorities and filter by DimDate.CalendarDate, which comes from the publisher's Data Date. Each source interval is one hour ending at UTCHourEnd. Use UTC for uniqueness and simultaneous cross-area comparisons. Use source local hour labels and operating dates for local profiles. Report the source snapshot and eligible count beside every result.

Two processes with shared dimensions

The target is a PostgreSQL fact constellation. Each fact keeps its own observation grain.

Table	Type and grain
FactHourlyDemand	Hourly Periodic Snapshot Fact. One selected authority and one UTC hour-ending instant with its demand, forecast and total net generation.
FactHourlyGeneration	Hourly Periodic Snapshot Fact. One selected authority, UTC hour-ending instant and original fuel category with a numeric reported value.

Dimension	Purpose / roll-up
DimDate	Operating date to month, quarter and calendar year. ISO week is paired with ISO week year. Weekday and season support profiles.
DimHour	Local hour-ending label 00-23. UTC distinguishes repeated clock labels.
DimBalancingAuthority	Authority code and name to current EIA region. One region member is not a duplicate fact population.
DimEnergySource	Original reported source header to reviewed SourceGroup. Preserve combined legacy categories.
DimSourceSnapshot	Source filename, complete-file hash and extraction time for lineage.

FactHourlyDemand is unique on authority and UTC hour. FactHourlyGeneration adds energy source to that grain. Aggregate generation to authority and UTC hour before joining to demand. For a group comparison, aggregate each fact independently to the same dimensions and then combine. One shared authority-hour with eight fuel categories still has one demand observation.

All dimensions and facts use warehouse-generated surrogate primary keys. Natural keys remain unique for reloads. Unknown identity or time rows are held for review. An Other/unknown fuel category is a valid published classification and remains distinct from a missing fuel value.

Demand and forecast definitions

Use explicit units, a consistent eligible population and denominators that can be inspected.

Metric	Definition and eligibility
M01 Observed demand energy	Sum ActualDemandMW x 1 hour over numeric, nonnegative actual observations. State coverage. Unit MWh. This is observed energy when hours have missing measures.
M02 Mean hourly demand	Sum eligible ActualDemandMW / number of eligible observations. Numeric nonnegative observations qualify. Group by local hour, operating date attributes and authority. Unit MW.
M03 Peak hourly demand	Maximum eligible ActualDemandMW within each authority and operating-month slice. Return every tied UTC instant with its local label. Unit MW.
M04 Forecast error and bias	Eligible pairs have numeric ActualDemandMW > 0 and ForecastDemandMW >= 0. Error = actual minus forecast. Bias = sum(error) / eligible pair count. Positive bias means underforecasting. Unit MW.
M05 MAE	Sum(abs(actual - forecast)) / eligible pair count, using the M04 population. Unit MW. Return N/A when no pair qualifies.
M06 WAPE	100 x sum(abs(actual - forecast)) / sum(actual), using exactly the M04 population. Return N/A for zero denominator. Aggregate from the sums, never by averaging area percentages.
M07 Underforecast frequency	100 x count(actual > forecast) / eligible pair count. Equal values remain in the denominator. Use M04 eligibility.

The acquired bulk headers specify MW. Store those reported hourly-average power values in MW. Compute interval energy as MW multiplied by one hour and report time totals in MWh. Average or maximize MW across hours. Sum power across distinct authorities only at aligned UTC instants.

Generation, shares and completeness

Signed net generation and source-category changes require explicit interpretation.

Metric	Definition and eligibility
M08 Reported net generation energy	Sum numeric signed GenerationMW x 1 hour by original source category or mapped SourceGroup. Retain negative values and indicate coverage. Unit MWh.
M09 Wind and solar category share	100 x sum(wind and solar GenerationMW x 1 hour) / sum(TotalGenerationMW x 1 hour). Use matched authority-hours with observed wind and solar categories, positive total generation and an available-fuel residual within max(10 MW, 1% of abs(total)). Show retained/excluded hours and category versions. Missing components are never silently set to zero.
M10 High-demand hours	Within each authority and operating month, compute the 90th percentile of numeric positive actual demand. Flag actual >= percentile, retaining ties. Compare source mix in that set with remaining eligible hours after generation is summarized at authority-hour grain.
M11 Missing / invalid measure rate	100 x affected expected authority-hours / expected authority-hours in the selected coverage window. Diagnose absent source rows separately from present rows with null or invalid measures. The frozen baseline's expected grid is 87,720 hours.
M12 Available-fuel reconciliation residual	Sum numeric reported fuel values at authority-hour minus TotalGenerationMW. Compare only when the total and at least one fuel value exist. Flag abs(residual) > max(10 MW, 1% of abs(total)). The threshold is a project diagnostic, not an EIA rule. A small residual does not prove every fuel category was reported.

Preserve exact reported fuel headers as category natural keys. The first 2024 file has nine fuel headers, while the next three have sixteen. Across the baseline there are nineteen distinct header meanings and sixteen with numeric observations. Roll up related solar and wind categories only through the documented mapping. The legacy combined hydro/pumped-storage category remains explicit.

Keep blank measures as null. Keep verified zero as zero, while excluding nonpositive actual demand from forecast scoring. A missing fuel column means the category is absent from that file's schema; a blank cell means no value was supplied. Neither proves zero generation. Unresolvable identity or timestamp rows enter the exception queue.

From source evidence to implementation

The source acquisition step is complete. The following warehouse flow is a design for the next phase.

Stage	Planned responsibility
Source files	Frozen official CSVs, reference workbook, selected extracts, checksums and source-row locators.
project_staging	Land text values and source metadata. Validate keys, times, numeric conversion and header changes.
project_dw	Load conformed dimensions, resolve surrogate keys and populate the two facts at their declared grains.
project_reporting	Publish grain-safe demand, forecast, generation and quality views for Power BI.

For the first implementation load, process the four frozen baseline files. Later, re-fetch the current six-month file into a new extraction folder and compare its hash. If changed, process the changed partition by business key. Recheck older files periodically for corrections. The script supports selecting a year/half and refreshing downloads; warehouse upserts and load watermarks are Phase 2 work.

Phase 1 activity	Completed output
Business analysis	Problem, scope, stakeholders, ten questions and measurable requirements.
Source validation	Four full downloads, saved subsets, reference workbook, manifests and repeatable profiling.
Dimensional design	Conceptual, logical and physical diagrams, grains, keys and a column dictionary.
Review	Requirement mapping, consistency checks and rendered-page inspection.

PostgreSQL is the target specified in the supplied Phase 2 brief. Python handles acquisition and later ETL; Draw.io/DBML provide editable diagrams. The designed calendar has 731 real dates, so the calendar alone exceeds the later brief’s stated 100 dimension-record minimum. Fact candidates exceed 10,000. These are source/design counts, not implementation results.

Source boundaries and traceability

Public URLs and included evidence let the reviewer reproduce the design assumptions.

Source	Reference
S1 EIA-930 bulk balance CSVs	https://www.eia.gov/electricity/gridmonitor/ Four files for 2024 and 2025. Exact URLs, retrieval times and complete-file SHA256 values are in source_manifest.json.
S2 EIA-930 reference workbook	https://www.eia.gov/electricity/930-content/EIA930_Reference_Tables.xlsx BAs and Regions tabs supply names and region labels. Frozen workbook included.
S3 EIA source definitions	https://www.eia.gov/electricity/930-content/about.html Time conventions, preliminary values, revisions and source-reporting limitations. Accessed 7 November 2026.
S4 EIA API documentation	https://www.eia.gov/opendata/documentation.php Optional later acquisition route. Bulk extraction is the implemented Phase 1 path.
S5 Instructor Phase 1 brief	DWBI Lab Phase 1.pdf, pages 1-3 Seven deliverables, process/grain/dimension tasks and the 30-mark rubric.
S6 Instructor Phase 2 brief	DWBI Lab Phase 2.pdf, pages 1-3 Compatibility only: PostgreSQL, staging/DW/reporting schemas and stated minimum row counts.

Constraints

Publisher observations are preliminary and may be revised. Reference labels are current labels, while the pilot dates are historical. Regional names do not imply that the selected authorities cover an entire region. Wind/solar category shares depend on reporting definitions and matched-hour coverage. Negative net generation can occur and is retained for review.

Deliverables are the proposal, business requirements, ten business questions, fact/dimension identification, grain definitions, schema atlas and Excel dictionary. The accompanying data package contains acquisition code, source extracts and a measured profile. Privacy exposure is limited to public aggregate operating data.